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Data Our Bodies Tell: Towards Critical Feminist Action in Fertility and Period Tracking Applications

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ABSTRACT

This article situates reproductive applications as an emerging “do-it-yourself” health technology in need of feminist technical communication action. The authors focus on Glow, a fertility and period tracking application, and argue that though this application promises user’s self-empowerment over their reproductive health, individual agency is often reduced. The authors consider how technical communication scholars can intervene in fertility and period tracking applications through a redesign of how consent is obtained when collecting user’s personal health information.

KEYWORDS

Fertility tracking application; feminist action; health technologies; reproduction; agency; consent

Introduction: situating feminist action in technical communication

Feminist perspectives in technical and professional communication (TPC) have, and continue, to expand the field’s scope and focus. For example, Thompson and Smith (2006) traced the history of feminist scholarship within the field and reported on a range of feminist contributions to TPC. Their report found that the bridging of TPC with feminisms opened up new scenes of inquiry regarding experiences of gender in TPC fields (Herrick, 1999; Ingram & Parker, 2002; Lind, 1999), gender bias in texts (Durack, 1998; Ranney, 2000), the recovery of women’s contribution to TPC (Flynn, 1997; Tebeaux, 1998), critiques of representations of women in TPC contexts (Cook, 2000; Flynn, 1997; Koerber, 2000; Petit, 2001), and the relationship between technologies and women’s bodies (Durack, 1997; Gurak & Bayer, 1994). Linking feminism to conversations in TPC has also made space for more critical and cultural approaches in the field (Scott, Longo, & Wills, 2007).

Despite these contributions, a more recent review of the literature on feminism and TPC sights an explicit decline in this work. Petersen and Walton (2018) find that “unfortunately, with a few notable exceptions (e.g. the work of Erin Frost, Kristen Moore, Dawn Opel, and Emily January Petersen), there has been a marked decline in TPC feminist research since the 1990s surge” (p. 5). In spite of the decline, they remind readers that “women continue to face gendered challenges that TPC scholars and teachers are well positioned to address” (Petersen & Walton, 2018, p.5), adding that, given TPC’s social justice turn, we must engage in critical action by critically analyzing for change. Bridging analysis with action, Petersen and Walton call in their article for a renewed “recognition of ways in which feminisms have, do, and can inform social justice work in TPC” (p. 3).

In this article, we take up and respond to Petersen and Walton’s (2018) call to bridge together feminist and social justice commitments in TPC by applying Royster and Kirsch’s (2012) *critical imagination* to identify new scenes of feminist action. As a feminist strategy, critical imagination works well to examine the location of feminist knowledge by encouraging us to look “in places at which we have not looked seriously or methodologically before,” “at practices and conditions at which we have not looked closely enough,” and “at genres that we have not carefully considered” (Royster & Kirsch, 2012, p. 72). Coming together to coauthor this article serves as a feminist

enactment of critical imagination. For instance, on the surface, our areas of research appear distinct and separate: one of us focuses on surveillance, privacy studies, and data collection; meanwhile, the other centers upon reproductive rhetorics, infertility, and community health literacy. Yet, by critically imagining places of overlap and intersectionality, we arrive at a shared scene of feminist inquiry: fertility and period tracking applications.

Applying our feminist line of inquiry to technical communication, we focus our attention on how user agency is actualized in fertility and period tracking applications. We understand that fertility and period tracking applications are marketed towards attracting various users: these applications can be used by individuals seeking to control and protect themselves from becoming pregnant while other users may seek to become pregnant and, therefore, may use these applications to indicate ideal times to conceive. At their core, fertility and period tracking applications allow users access to their menstrual data to increase their reproductive health literacies. How one selects to use that information – whether to avoid pregnancy or attempt to become pregnant – is up to their personal discretion. We start by questioning how much agency users actually have over their reproductive health when using these applications and how we, as technical communication scholars and teachers, may take steps to enhance user interaction in these fertility and period tracking applications.

Our critical approach, which responds to research and pedagogy concerns, follows Jones's (2016) call to recast the field of TPC towards one that "make[s] a difference in the lived experiences of others" (p. 345). Because technology companies tend to proclaim their fertility and period tracking applications improve the reproductive health literacies of their users, these applications have the possibility to embody TPC's social justice aims. For example, these applications increase one's autonomy and access to personal health data, which can be easily downloaded onto one's mobile phone or tablet as a means to reduce barriers to reproductive and, more specifically, fertility care. In the context of these applications being used to try and conceive, fertility and period tracking applications can further reduce the need to see a fertility specialist, whose services are frequently not covered by insurance and can result in high out-of-pocket costs. Yet, though these applications can be appealing to users seeking to conceive, we must remember that, just like other technologies, these applications are "not neutral or objective" (Haas, 2012, p. 288). Feminist, social justice approaches to technical communication ask researchers in the field to consider facilitating more ethical, genuine relationships between these technologies and users. Such work can be done by embracing feminist pillars that include "celebration of difference, social change, acknowledgement of backgrounds and values, inclusion of women's experiences, study of gaps and silences, and new sources of knowledge" (Lay, 2004, p. 429).

These critical feminist, social justice, and cultural perspectives about technology prompted us to turn our attention to Glow, a popular fertility and period tracking application marketed to help women either avoid an unwanted pregnancy or assist women in becoming pregnant. In this article, we focus our attention on the latter use. Although many other fertility and period tracking applications exist, our focus on Glow is intentional. First, Glow was launched by PayPal founder Max Levchin and former Google executive Mike Huang. The launch of this application, particularly by two men with seemingly little expertise in reproductive health, interested us – especially as their positions in the technology industry increased the press surrounding this particular health application. Second, Glow offers an optional premium service called "Glow First." For \$50 per month, this premium service markets itself to couples trying to conceive, guaranteeing after 10 months of use a small "payout" to be redeemed at a local fertility clinic. To our knowledge, no other fertility and period tracking application has offered such a program. The strategic marketing in the launch of this application informs our decision to focus specifically on Glow, rather than other fertility and period tracking applications. Moreover, we find that fertility and period tracking applications, though rising in popularity and usage, have yet to be critically examined by feminist technical communicators. Given this gap, we examine how this application promises user self-empowerment at the expense of commodifying private health data. A critical feminist analysis of

the application's Terms of Service and Privacy Policy reveal a space for technical communicators and feminist rhetoricians to respond by redesigning how these applications obtain users' consent.

To support our claim, we begin by situating women's health applications, like fertility and period tracking applications, as a technical communication text. Later, we assemble a theoretical framework around rhetorical agency and empowerment to guide our analysis of fertility technologies and applications. Bridging critical analysis to evoke more socially just action, we critically analyze Glow through Frost's (2016) discussion of efficiency to point to gaps in which the application actually erodes user agency. In doing so, we identify areas for action to address these concerns and consider how technical communicators could assist in the redesign of an application's technical documents to improve user's agency. A series of implications are offered to feminist technical communication scholars interested in further taking up this work as potential users, instructors, and researchers.

Women's health applications as technical communication texts

We find that women's health applications are forms of "do-it-yourself" health technologies. These applications function as technical communication texts in two particular instances. One, they collect and interpret personal health data to enhance women's health literacies and agency in health-related decisions. Two, they are a widely accessible and affordable form of technology that support increasing eHealth and Health 2.0 trends (IMS Institute for Healthcare Informatics, 2015). For example, the widespread availability of these applications to mobile users via download enable an ease in accessing one's health information. Additionally, many of these applications cost little to no money. Glow, for example, is free to download and though it costs more for premium services, such as "Glow First," users of the free application still receive personal, customized information regarding their reproductive cycle. Many other fertility and period tracking applications are also of little cost – allowing for increased access to personal health information (Moglia, Nguyen, Chyjek, Chen, & Castaño, 2016).

Further, a 2013 study by ByteMobile reported that pregnancy applications outnumbered the number of fitness-focused applications (Dolan, 2013). Popularity of these applications continues to grow. A 2015 IMS Institute for Healthcare Informatics report found that, of the 90,088 health applications in the U.S., Apple iTunes store has 7% of them marketed solely as products assisting with women's health and pregnancy ("Patient Adoption"). Although many of these applications focus on tracking and monitoring the health of the pregnant body and the fetus, several of these applications are also focused on developing fertility awareness for their female users. In fact, many new health initiatives are researching the effectiveness of fertility applications as methods to either try to become pregnant *or* avoid an unwanted pregnancy.

The Healthy People 2020 campaign is one example of a current initiative conducting research on the effectiveness of fertility applications that increase fertility literacy to reduce unintended pregnancy. This campaign, supported by the U.S. Department of Health and Human Services, seeks to reduce unintended pregnancy by 10% by 2020 through evidence-based family planning programs that rely upon smartphone technologies and their mobile applications ("Healthy People," 2018). In these circumstances, fertility and period tracking applications are tools to increase fertility literacies in demographics traditionally viewed as "at risk" for an unwanted pregnancy in an effort to promote fertility awareness ("Healthy People," 2018). Simultaneously, women trying to become pregnant regularly rely on these applications. A 2015 study found that women who are younger than age 35, trying to become pregnant, and have regular cycles are more likely to use applications to track their periods than other women (Lanham & Christensen, 2015).

As we stated earlier, many of these fertility and period tracking applications are frequently described as empowering female users by providing seemingly quantitative health information through affordable and accessible methods. Although clinicians see these fertility applications as helping to increase communication between patient and provider, they also express caution in the accuracy of these applications. Specifically, clinicians working with young, female patients suggest

the need to discuss what applications their patients are using to encourage more critical evaluations of the applications that their patients use. Although clinicians recognize the benefits these applications provide in beginning conversations with their healthcare providers, they express a need for users to evaluate the motivations of the application's developers as well as how the application gathers users' privacy and health information (Haelle, 2018).

Given how these applications are becoming more commonplace and offer a supplement to improving healthcare, we find them as an emerging technology in need of a feminist technical communication lens. Further, as these applications seek to better their users' health, we see them as supporting the bridging of feminist approaches to social justice work (Walton & Jones, 2013). In the section that follows, we bridge feminist approaches with social justice commitments by assembling a theoretical frame to critically understand the operationalization of users' agency and empowerment with these applications.

Agency and empowerment in fertility technologies

Reviewing literature in TPC, we draw primarily on Jones's (2017) theoretical framework of agency and empowerment to situate it in conversation with fertility technologies that, often when marketed, promise increased individual agency over their reproductive health. Although Jones (2017) focuses on Black entrepreneurship, we find her discussion of agency and empowerment applicable to understand how increasingly popular fertility technologies operate. Many contemporary fertility technologies seek to increase patient agency and sense of control over reproductive health. We critique the promise of increased agency in these applications by focusing on the relationship between agency and empowerment. Jones's (2017) work suggests that empowerment more accurately captures the process for creating moments of agency. Additionally, her work enables us to critically analyze how (and if) period and fertility tracking applications act as empowering technologies that offer users increased agency over their fertility.

Jones (2017) begins her careful discussion of agency by referencing Koerber's (2001, 2006) scholarship on rhetorical agency. For Koerber (2006), rhetorical agency is "negotiation among competing alternative discourses" (p. 94). Such a definition implies that:

rhetorical agency requires two components: (a) an awareness of the rhetorical situation, including exigency, kairos, and an understanding of existing discourses or arguments, and (b) the ability, opportunity, or rhetorical space to act. In this sense, rhetorical agency is more about possibility than outcome. (Jones, 2017, p. 325)

Jones's (2017) use of Koerber's definition of rhetorical agency suggests that it "is more about possibility than outcome" (p. 325). This understanding of agency implies a need to take up how agency turns into action. Drawing on cultural empowerment theories (Adams, 1996; Cushman, 1996; Jabareen, 2009), Jones articulates a theoretical framework that argues: "while rhetorical agency pushes us to think about possibilities for action, cultural empowerment provides a framework for considering both process and outcome" (p. 325). By repositioning empowerment as a process that leads to and facilitates action, Jones emphasizes that "while rhetorical agency creates a (negotiated) rhetorical space for resistance and potential change, empowerment represents the actual enactment of rhetorical agency in a transformative way" (p. 342). Her attentive care to how agency itself does not inherently lead to resistance and self-empowerment is important particularly in the context of contemporary fertility and period tracking technologies, which are frequently marketed via a cultural discourse that claims to enhance female agency over their fertility.

An example connecting Jones's critical approach to rhetorical agency and empowerment to fertility technologies is Frost and Haas (2017) examination of fetal ultrasounds. Although their work does not explicitly mention Jones's (2017) theoretical frame, much of their article supports Jones's arguments about agency and empowerment and links that framework to scenes of fertility and reproductive technologies. Specifically, Frost and Haas offer a case study on the fetal ultrasound so as to interrogate "the potential for hegemonic uptakes to undermine women's agency over our

bodies” (p. 89) and point to prior scholarship (Georges, 1996; Hubbard, 2003; Mitchell, 2001) that has suggested that “the visibility of pregnancy posit that a rhetoric of increased trivialization of pregnancy has contributed to a reduction of agency for pregnant women” (p. 95). For them, the fetal ultrasound marks a moment in which physiology and science erode women’s agency in the pregnancy process. In fact, too often fertility technologies are uncritically celebrated as liberating technologies whereby, “women are indoctrinated into a culture that expects us to look outside of our bodies to assist us in determining and confirming what is happening inside our bodies” (Frost & Haas, 2017, p. 96). These technologies are made, then, to appear as if they provide moments to exercise agency but, in fact, are reinforcing colonizing reproductive medical structures.

For example, Frost and Haas’s (2017) case study points to multiple scenes in which technologies actually erode women’s agency, including the clinic whereby the “physicians, sonographers, nurses – tend to embody the most power and exercise the most agency in defining, prescribing, diagnosing, and describing embodied pregnancy and infertility experiences...[when they] do not have the[se] embodied experiences” (p. 99). They call for feminist technical communicators, rhetoricians, and women in general to adopt a more critical, decolonial approach to reproductive technologies that promise increased agency over one’s fertility and health, suggesting that “patients have agency in the ultrasound process, from seeing the screen to bringing a companion for moral support and asking questions of the sonographer” (Frost & Haas, 2017, p. 100). Their critical analysis of the fetal ultrasound is helpful in situating a feminist critique of reproductive health technologies alongside Jones’s critique of agency. Although fertility and reproductive technologies like the ultrasound might promote a sense of women “doing good” by “taking care of” their impending child, they also establish norms that we can critique. Without critique, we miss how these practices all too often surveil the female body, erode agency, and – ultimately – disempower women.

For the purpose of our analysis, we build on Jones’s (2017) and Frost and Haas’ (2017) critical feminist and social justice work in technical communication, finding that fertility and period tracking applications affect agency in a similar manner as other fertility technologies, like the fetal ultrasound. Although these applications seek to build fertility awareness and, hence, improve individual agency and control over one’s own fertility, the process by which this is done often erodes the actual empowering outcome. As feminist technical communicators and rhetoricians who believe in developing technologies to increase female agency, we see a need to develop a framework to critically analyze these technologies so as to inform practices that invite feminist action.

Assembling a critical feminist analytical framework

Returning to Petersen and Walton’s (2018) call to bridge feminist critical analysis with social justice action in TPC, we build a framework to critically analyze Glow with regard to agency and empowerment. Petersen and Walton (2018) contend that though “action is needed to redress inequities,” they “also see a potential danger in the field’s shift toward critical action if that shift is not carefully informed by critical analysis” (p. 3). Understanding that analysis and action are necessary to move forward critical frameworks to best support the potential of TPC social justice scholarship, we assemble a critical feminist analysis of Glow to, later, offer an action-oriented response with implications for technical communication.

Frost’s (2016) apparent feminism methodology is useful for building a critical analysis because it “can help us think about the diversity of potential audiences, what productive or efficient work is, and how to do our work in socially just ways” (p. 11). Moreover, her framework offers a model for engaging in feminist, social justice analysis, which can be applied toward understanding how agency and empowerment are operationalized in Glow. Frost asserts that, “technical communication scholars and practitioners need an apparent feminist methodology in order to intervene in situations in which technical documentation unfairly and uncritically engages in oppression while feigning objectivity” (p. 4). Applying apparent feminism to Glow allows us to identify

moments for feminist action by redesigning how individual reproductive health agency in Glow is actualized while using the application.

We focus on the third bullet to Frost's methodology: efficiency. Explaining her methodology, Frost claims that efficiency "is commonly understood as the balancing point at which we achieve the best result while expending the least amount of energy" and argues that though technical communication is concerned with issues of efficiency, it is too often "most focused on the latter part of that definition" (p. 16). For efficiency to support feminist aims, however, technical communicators must also consider the results that are produced and who those results benefit. Frost thus offers a revised definition of *efficiency* suggesting that we focus "primarily on audiences as a component of best results" (p. 16). This understanding means that, in the context of apparent feminism, "efficiency can be used as an ethical end, but only if we understand efficiency as being focused primarily on effects on human lives and secondarily on speed and energy" (p. 19). Fertility applications, like Glow, are efficient technologies that seek to support the end result of their users, namely to either become pregnant or avoid pregnancy, but how these results are achieved demands further investigation.

Frost reminds us that "understanding efficiency as something that benefits those already in power, something that reinforces the status quo...prevents us from doing good, ethical, socially just work" (p. 20). Next, we assess the efficiency of Glow through Frost's apparent feminism methodology to measure the extent of agency the app provides. To do this, we critically analyze how the application works and operates to produce information to its users. Such analysis aids an understanding of how use is consented to through technical documents like the Terms of Service (ToS) and Privacy Policy (PP). Analyzing the design of the application as well as its corresponding legal documents are two areas ripe for feminist technical communication intervention.

Glow: a fertility and period tracking application

In this section, we contextualize Glow as an application marketed to users encountering trouble conceiving and situate the application within the fertility treatment market – noting how it is marketed, often, to a healthcare demographic that cannot afford fertility treatments. Next, we provide a description of how these users interact with the application. Lastly, we turn our attention to how users engage with Glow so as to examine the application from the lens of efficiency. In doing so, we hope readers understand the appeal of Glow to the infertility community.

Infertility & Glow

Fertility treatment is a major and often under-recognized health crisis, with the Centers for Disease Control and Prevention (2016) reporting that one in eight couples in the U.S. will receive an infertility diagnosis in their lifetime. Skeptics critiquing infertility as a health crisis point to a series of medical advances developed over the last 40 years to combat infertility. Yet many of these critiques fail to account for the increasing out-of-pocket costs required to pay for fertility treatments. In fact, in 2018, only 16 states in the U.S. mandated full or partial insurance coverage of fertility-related treatment ("Infertility Coverage by State," 2018). This statistic means that if a person lives in a state that does not mandate insurances to cover fertility treatment, or if they do not have insurance, paying out-of-pocket for one round of in vitro fertilization (IVF) equates (on average) to a cost of \$20,000. Recently, Smith, Tilling, Nelson, and Lawlor (2015) reported that two thirds of patients tend to undergo six rounds of IVF before giving birth to a living infant. These findings illustrate the extensive financial burden of living with infertility, not to mention a host of emotional stresses that come with both figuring out how to pay for treatment and undergoing treatment – possibly for multiple rounds.

Therefore, free fertility applications often appeal to many in the infertility community as an affordable fertility tool. Glow is described on the Apple Store Preview Page as a free app "designed to help you take control of your reproductive health" (Terms of service 2016) and further targets

individual users seeking to conceive by boasting on its website. Glow's site also advertises the app by saying that "women are 40% more likely to conceive when using Glow as a fertility tracker" (2016). Situated in the context of infertility, Glow markets itself as an alternative to more costly forms of treatment, addressing two current gaps existing amongst infertility service providers: (1) the ability to collect and interpret bodily data from the privacy of one's own home, without the consultation of a fertility specialist; and (2) the ability to use the app at no cost and join and communicate via forums with a community of other individuals using the application.

User interaction with Glow

Assessing the user's interaction with Glow helps to enact a more critical approach to how Glow impacts user agency. From its marketing and stated intent to help those struggling to conceive, Glow clearly fills current gaps in fertility healthcare. Yet, as a technology reliant upon collecting its users' personalized health data, we consider how it promotes a rhetoric of efficiency antithetical to Frost's (2016) definition: focused too much on delivering quick, analytical data without a more ethical consideration of how its processes actually erode individual user agency.

Examining the step-by-step process users undergo to enroll in Glow, we situate Glow as an application that should take more explicit efforts to make clear when and how users give up their data to Glow at the expense of possibly becoming pregnant. Applying Jones's (2017) work on agency and empowerment, we find Glow does not fully empower its users by its opaque consent process to obtain access to the user's personal health information. In what follows, we apply an autoethnographic (via screencaptures of the application during use) approach to user engagement with the application and detail the steps an individual must undergo in order to create a user account with Glow.

Step 1: The user downloads the application, either through the Apple Store or Google Play. It is this first step where users consent to the ToS and PP. (see [Figure 1](#)).

Step 2: The user consents to "get started" with the app by clicking on the purple "Get started!" button. (see [Figure 2](#)).

Step 3: The user must select their desired objective in using the application, either to conceive, avoid pregnancy, or as a male user to link his data with a female partner.¹ (see [Figure 3](#)).

Step 4: The user enters personal information to create an account. (see [Figure 4](#)).

Step 5: The user indicates the length of time they have been trying to conceive. (see [Figure 5](#)).

Step 6: The user must disclose the date of their last period. (see [Figure 6](#)).

Step 7: The user must disclose their average menstrual cycle length. (see [Figure 7](#)).

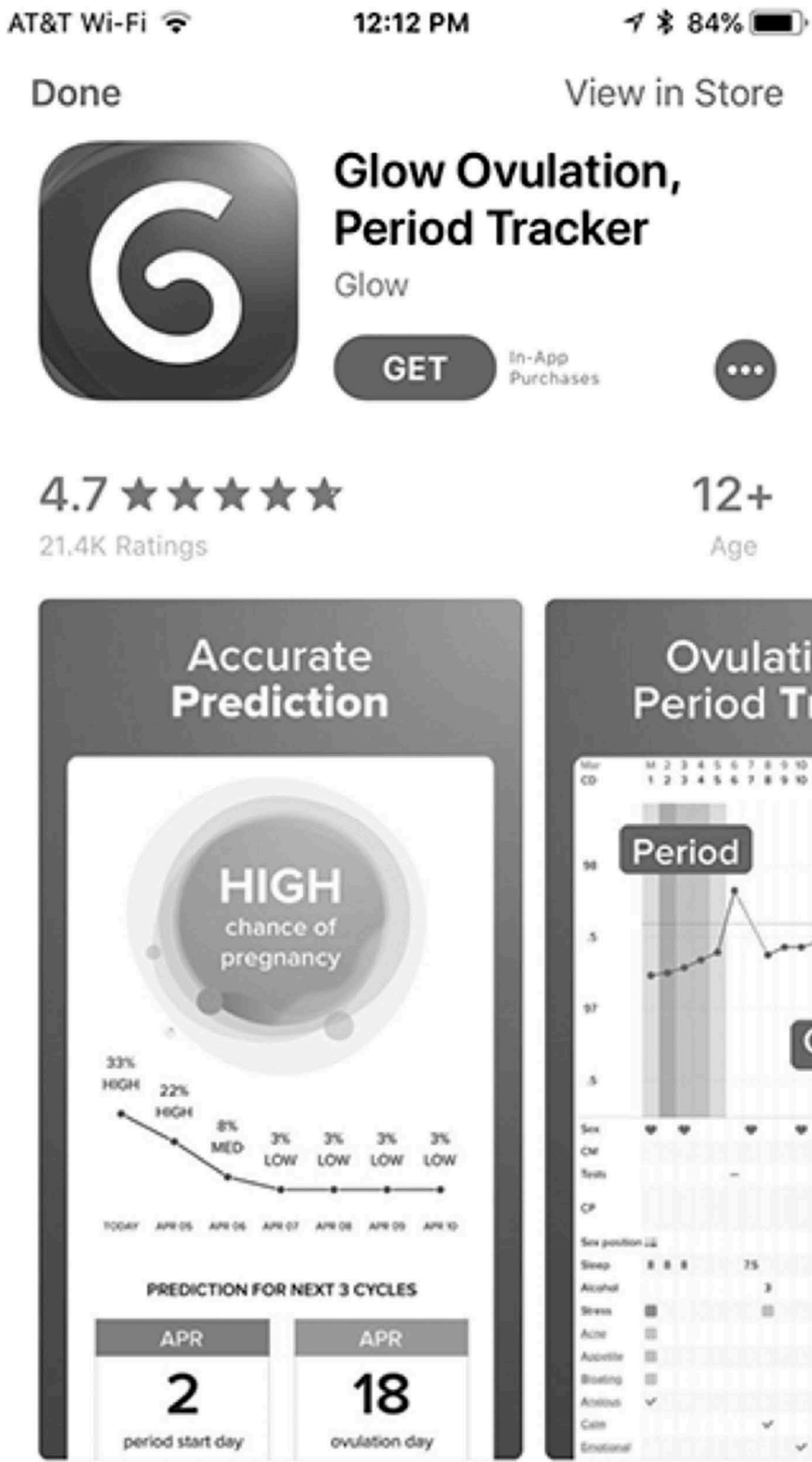
Step 8: After entering this information, the application generates a calendar that indicates when ovulation likely occurred and when the user's next period will most likely arrive, based on the information collection in Steps 5–7. (see [Figure 8](#)).

Step 9: Glow invites the user to receive personalized notifications concerning the user's best time to conceive based upon the data that they enter in the application. (see [Figure 9](#)).

Step 10: The user is notified about how likely they are to get pregnant without any additional information about this "diagnosis," which points toward a need for future analysis that can critique how the app's interface design may or may not support user agency as well. (see [Figure 10](#)).

Step 11: The user is finally granted access to the app interface, shown below, and offered the opportunity to purchase medications to increase the chances of pregnancy directly through the application. (see [Figure 11](#)).

Step 12: The user is notified to continue adding data to the app and complete their health profile to increase accuracy of the app. (see [Figures 12 and 13](#)).



Glow®

Provides
Accurate cycle predictions

Get started!

Log in



Figure 2. Screen capture of the first screen of the Glow application once downloaded.

A critical analysis of Glow

If technical efficiency should be understood as not only meeting its intended outcome, but also as also serving people affected by placing “primary importance on the human rights of stakeholders”

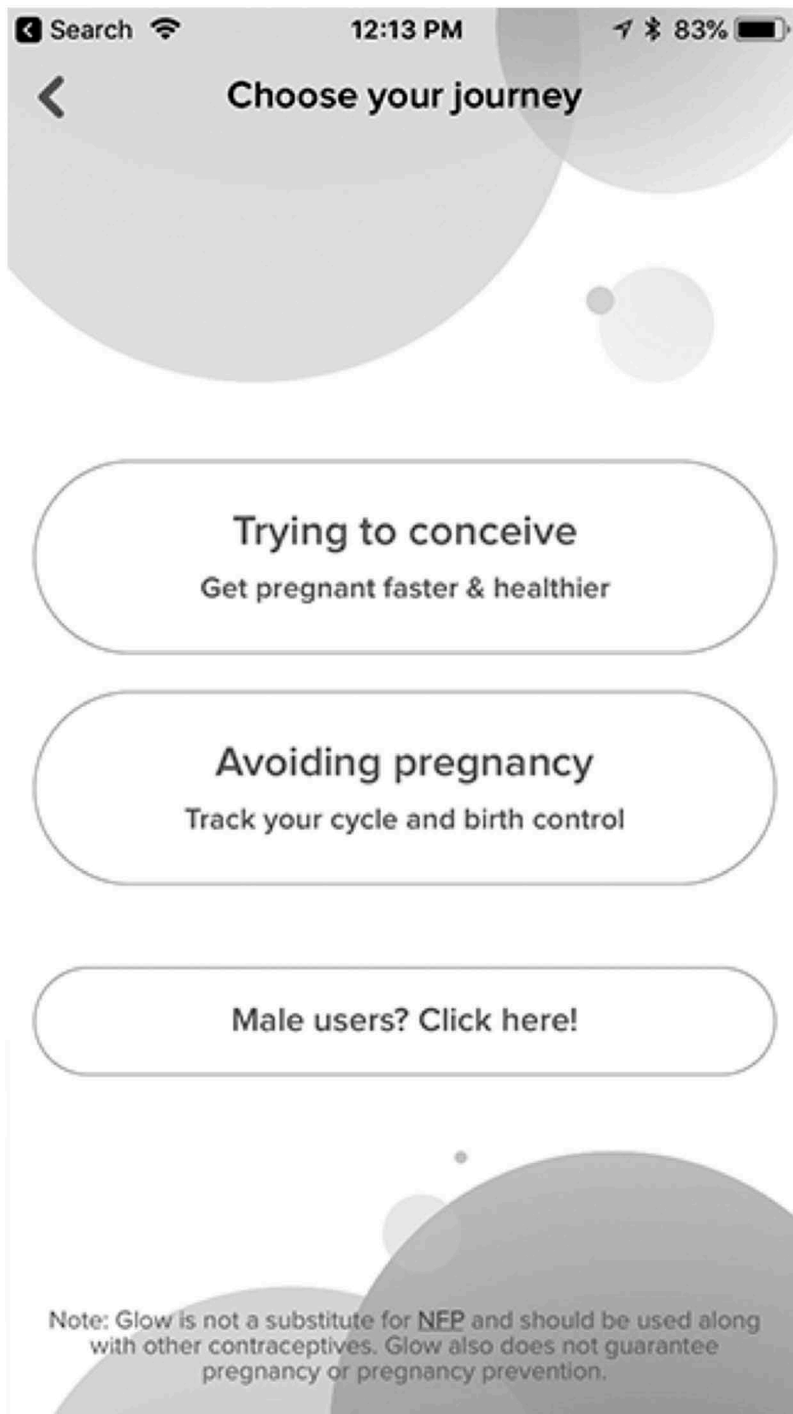


Figure 3. Screen capture of the second screen of the Glow application once a user has agreed to "Get Started" or log in to the app.

(Frost, 2016, p. 20), we must analyze how Glow treats the individual rights users have to their personal information when using/consuming the application. Specifically, we raise concern about how the application fails to make the collection and use of personal data transparent to its users.

Search 12:13 PM 83%

< Welcome

Full name First & Last name

Email name@example.com

Password 6 characters

Birthday Required

By continuing, you agree to our [Terms](#) and [Privacy Policy](#)

Next

Q W E R T Y U I O P

A S D F G H J K L

↑ Z X C V B N M ↵

123 😊 🎤 space return

Figure 4. A screen capture of Glow’s log-in screen that asks a user to input their full name, an email address, a password, and their birthday. All of this information is required and determines that the user agrees to Glow’s Terms and Privacy Policy.

Such attention is further informed from Jones’s (2017) theoretical framework on agency and empowerment, arguing that scholars need to pay “attention to how rhetorical agency is enacted” (p. 325).

The screenshot shows a mobile application interface. At the top, there is a status bar with a search icon, signal strength, time (12:14 PM), and battery level (82%). Below the status bar is a navigation bar with a back arrow, a progress indicator (three dots, the first is filled), and a search icon. The main content area has a large heading "How long have you been trying to conceive?" followed by a subtext "Glow will customize your fertility insights based on this information." Below this is a scrollable list of options: "4", "5", "(Choose) weeks", "6 months", and "7 years". The "5" option is currently selected. At the bottom of the screen is a large grey button labeled "Next".

Search 12:14 PM 82%

< ● ● ●

How long have you been trying to conceive?

Glow will customize your fertility insights based on this information.

4

5

(Choose) weeks

6 months

7 years

Next

Figure 5. Screen capture of the first question the Glow application asks a user: how long have you been trying to conceive?

We question how agency is enacted throughout the process of using the application and not just as an outcome measured through the user's end result of either achieving or not achieving a pregnancy: Are users positioned as agents through all stages of their interaction with Glow? We



Figure 6. Screen capture of the second question asked of users in the Glow application: Do you know the date of your last period? The question precedes a calendar that requires the user to click on a date.

argue no. For instance, in [Figure 10](#), the Glow interface sends a notification that this user has a “medium chance at pregnancy today” without offering any additional information or suggestion actions. Because of this lack of information, users are left to make their own, possibly uninformed, decisions about their reproductive health in that moment. We find that, in the application’s

The screenshot shows a mobile application interface. At the top, there is a status bar with a search icon, signal strength, time (12:15 PM), and battery level (82%). Below the status bar is a navigation bar with a back arrow and a progress indicator (three dots, the middle one is filled). The main content area has a title "Do you know your average cycle length?" followed by a subtitle "How many days from the start of one period until the start of the next period? It's okay to guess." Below this is a list of numbers: 27, (Choose), 28 days, 29, and 30. The number 28 is highlighted with a thick border. At the bottom is a large dark gray button labeled "Next".

Search 12:15 PM 82%

< ● ● ●

Do you know your average cycle length?

How many days from the start of one period until the start of the next period? It's okay to guess.

27

(Choose)

28 days

29

30

Next

Figure 7. Screen capture of the third question asked of users in the Glow application: Do you know your average cycle length? Users should select a number of days up from 1-30.

interface, this ambiguity negatively impacts user agency. Ambiguous language operates beyond and behind the interface as well. In the section that follows, we turn to Glow's ToS and PP to address how Glow approaches personal health data of their users.

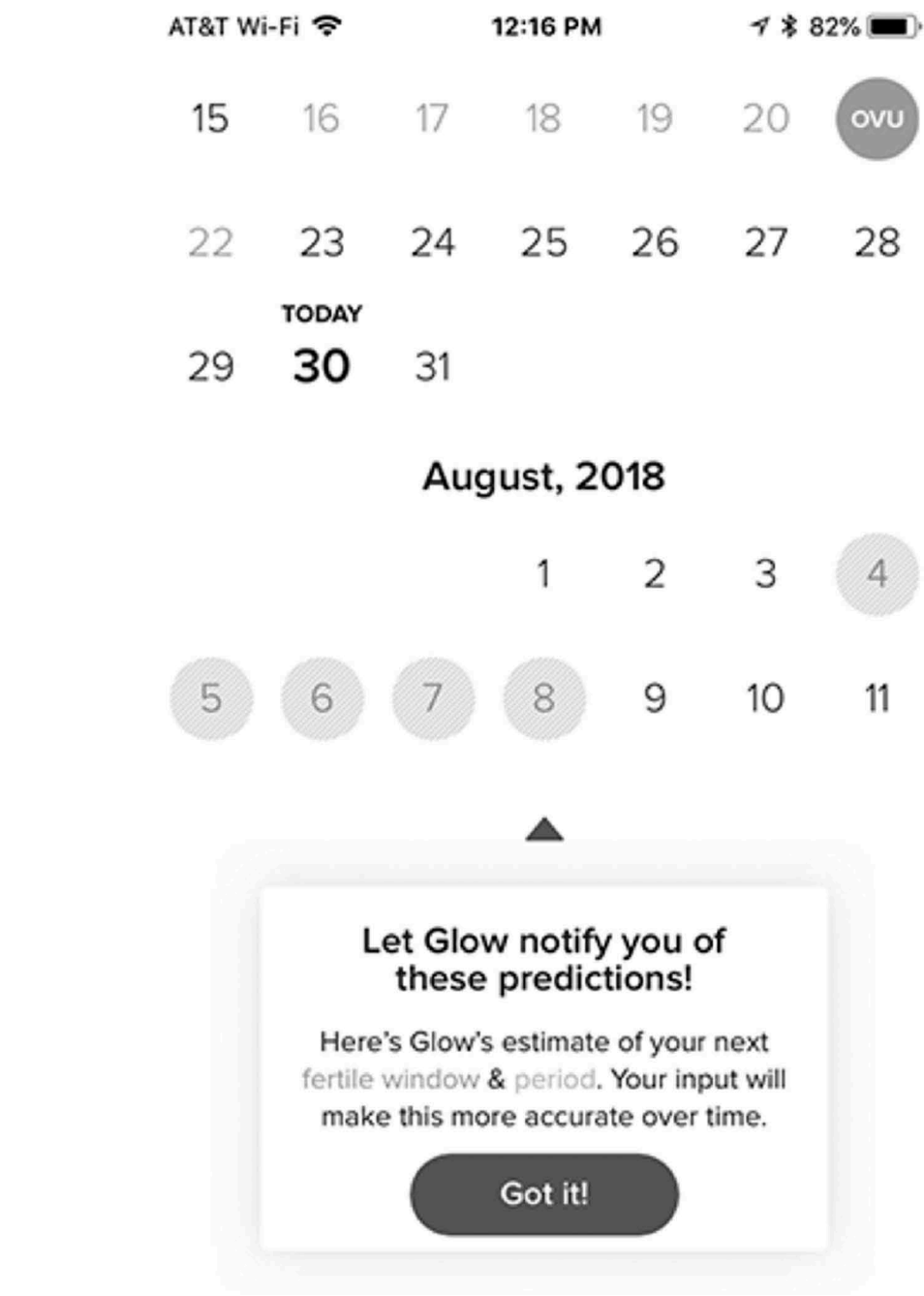


Figure 8. A screen capture of the Glow app's calculated prediction of the user's next ovulation window in a calendar. This information is presented once the user has completed the previous set of questions.

Critical analysis of how Glow obtains consent to access users' personal data

Technology companies tend to inform their users about data protection as well as the access and restrictions third-party affiliates may have to user data in their technical legal documents like the ToS/Conditions and PPs. Yet, like many digital platforms and technological applications, the interface of Glow fails to invite user interaction with these documents. Glow embeds its ToS and PP as

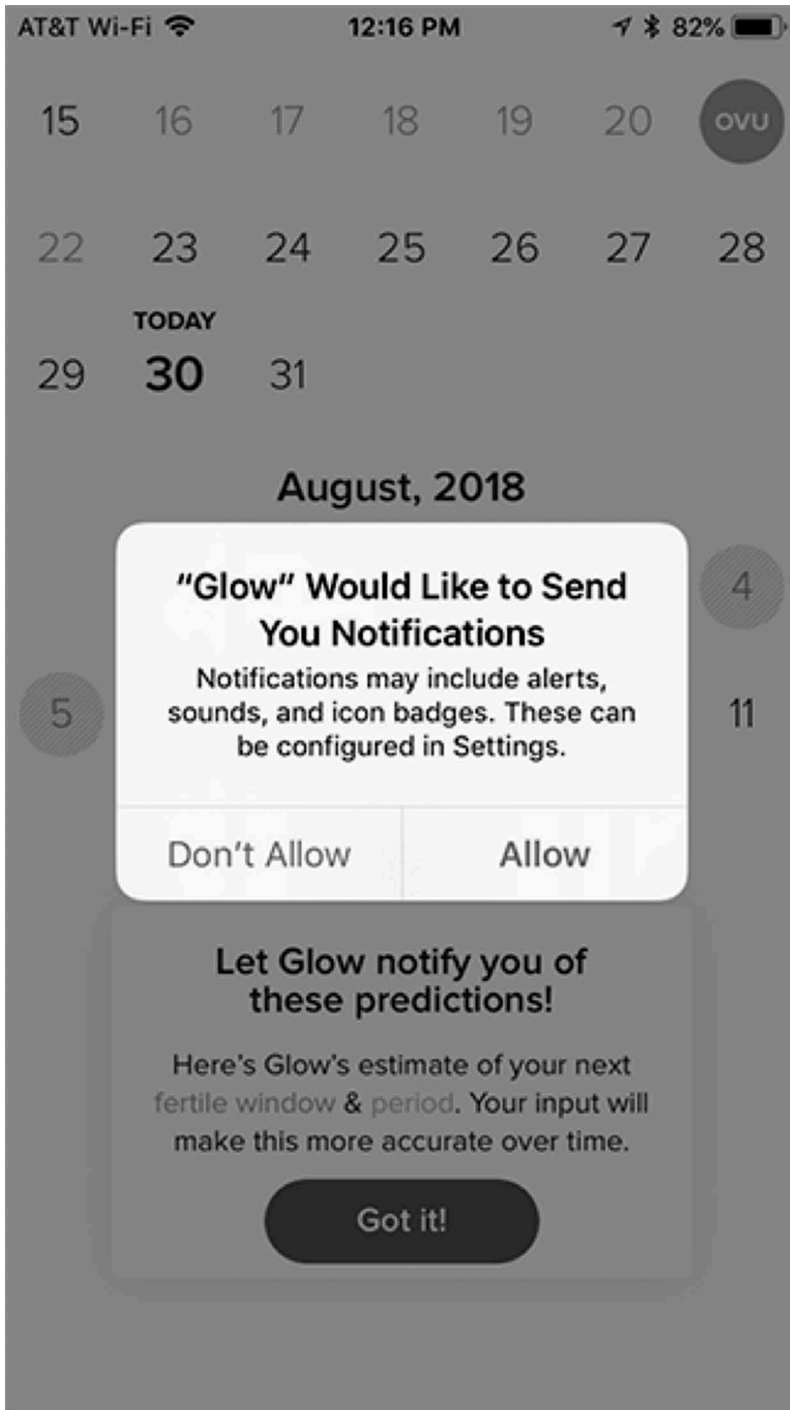


Figure 9. A screen capture of the Glow application requesting permission to send the user mobile notifications.

a part of the consent process to download the application (see [Figure 4](#)), meaning that when the individual proceeds to download the app through the Apple Store or Google Play, the user automatically (and we would argue often unwittingly) consents to the app's ToS and PP. Consent,

AT&T Wi-Fi

12:16 PM

82%



Hey [name], based on your info,
we calculated your chance of
pregnancy today.

Figure 10. A screen capture of the Glow application's calculation of the user's chance of getting pregnant on that date due to the personal information provided.

in this case, is visually embedded, and, we argue, therefore hidden.² The user is never required to read and/or interact with the consent agreements. Instead, consent is assumed when users agree to download the application. After this, consumers will immediately begin to submit all of the information in the app (as we outlined prior), having little-to-no knowledge where this information will go, who will have access to their information, and what profit will be made off of this information once it is made into data. As such, we question how consensual such a legal agreement actually is.

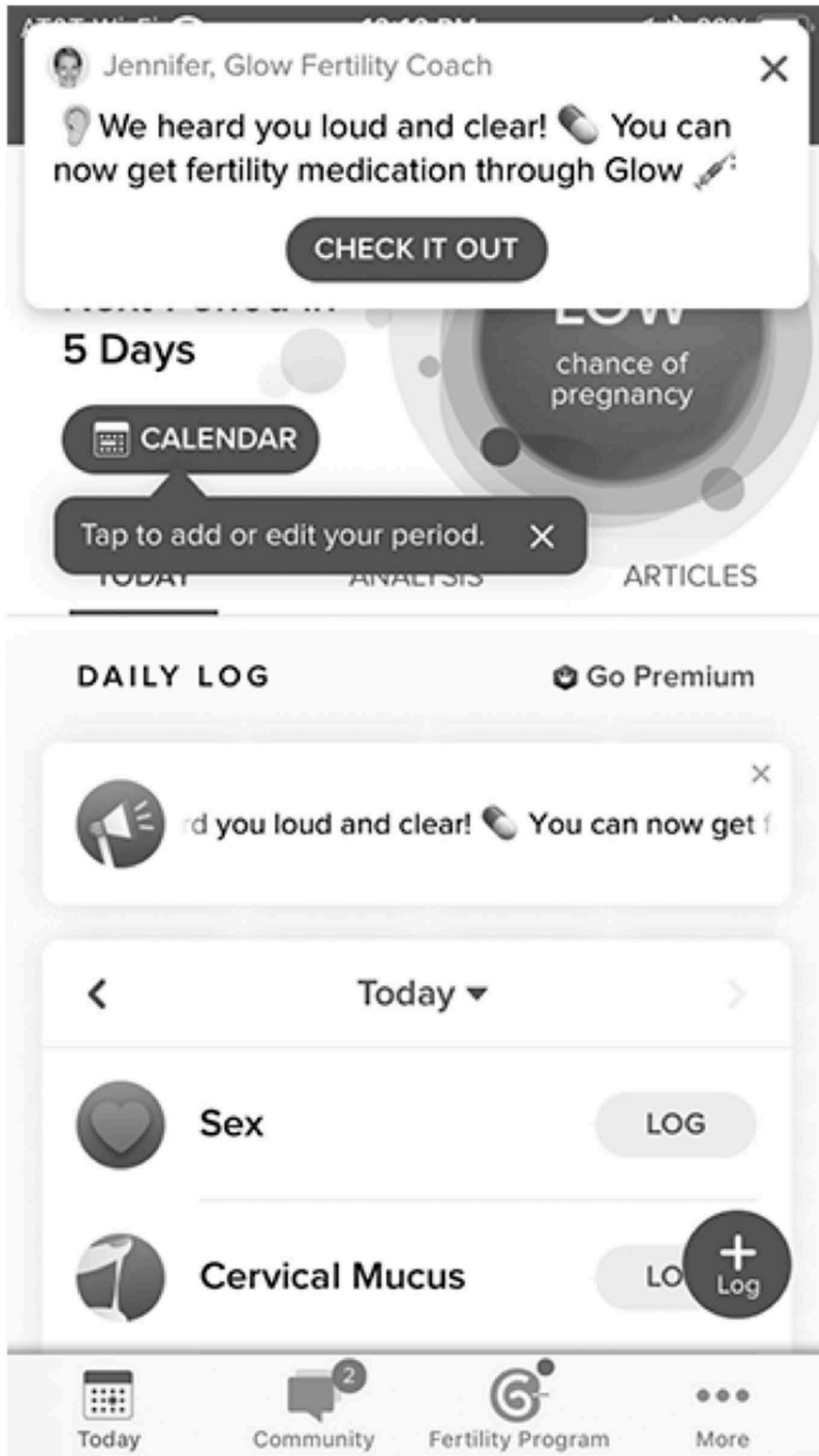


Figure 11. A screen capture of the Glow home screen once the user has inputted all the needed personal information. The image includes an advertisement for fertility medication.

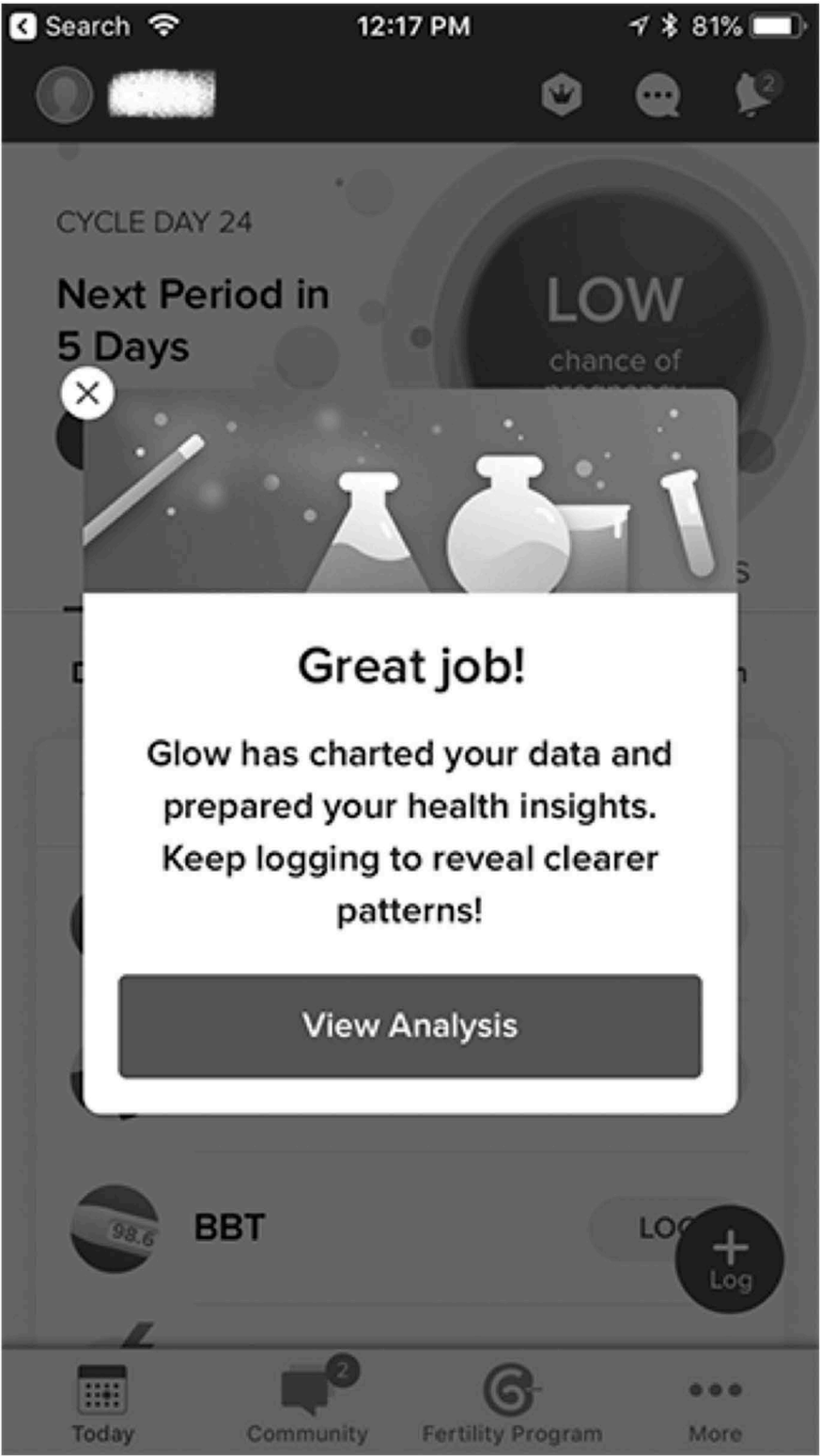


Figure 12. A screen capture of the Glow app providing an in-app update that notifies the user that their data has been compiled and asks them to continue logging in their data to receive "clearer patterns" from their data.

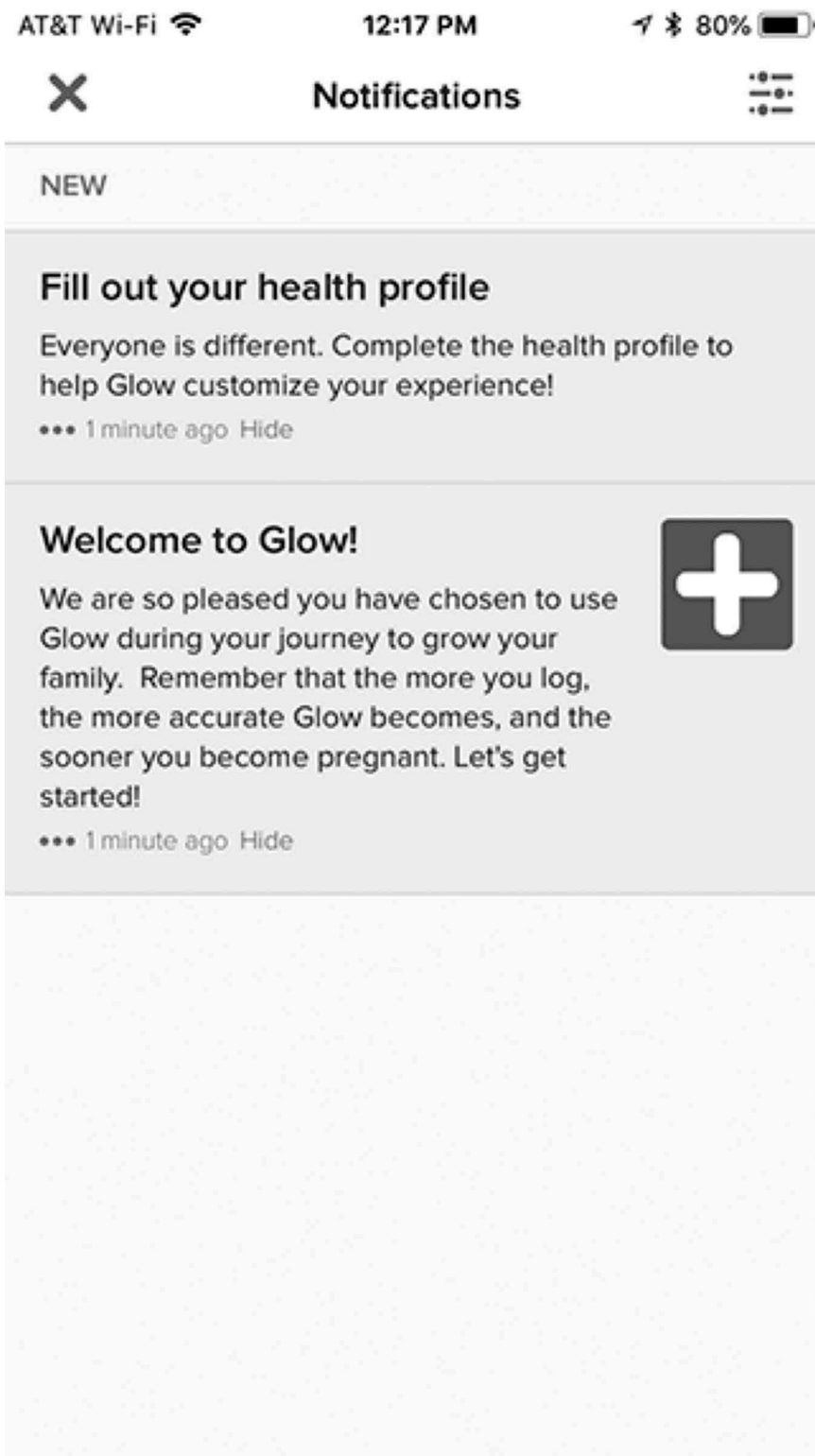


Figure 13. A screen capture of the user's notifications' screen in the Glow application.

Critical analysis of the Terms of Service

It should be made clear that the users signing up for Glow is never prompted to a page explaining how their personal data becomes Glow's property. If the user wants to review Glow's ToS, the user can find this document on Glow's website. Reviewing the ToS, Glow makes clear that Glow is the sole owner of the information it collects from the user, writing that "If you do not want us to use your data as described in our Privacy Policy, please do NOT use the Services" (2016). Figure 14 and the text highlighted within it further articulate Glow's stance on owning its users' data.

Glow's ToS expressly state that the company has the right to indefinitely retain and continue to own all user data collected, regardless of whether a user terminates their account or not. Glow mandates that users, after terminating their account, must "destroy and remove from all computers, and other storage media all copies of the any [sic] intellectual property owned by Glow" (2016). All intellectual property owned by Glow includes any content or information the user has posted through its service (including communication exchanges with other users in the message boards and private messages), all user data (such as period information and health updates), and even any and all emails exchanged between the user and the company.

Despite the application's claim to own the user's data, the ToS denounces any sort of liability to the loss of personal information or data, explaining that "Glow is not liable under any theory of law for any loss of information or data" ("Terms of Service," 2016). This is of concern, given a 2016 *Consumer Reports* finding that suggested they found several security and privacy flaws in Glow, enabling hackers to exploit these vulnerabilities to access users' personal health and financial information (Beilinson, 2016). Finally, the ToS clearly establishes that:

Glow makes no commitments about the reliability, availability, or security of the Services or their ability to predict your cycles and related transmission of sensitive data other than that it will use commercially reasonable efforts to protect your information as described in our Privacy Policy. ("Terms of Service," 2016)

Such a statement, particularly the phrase "commercially reasonable," we argue is vague and unclear. This is especially confusing given, as we will see in the PP, that Glow makes it clear it does not sell any user's personal information to a third party unless given permission by the user. As such, we are left to question: How is user data commodified by Glow for commercial purposes?

Critical analysis of the privacy policy

The ToS frequently references readers back to Glow's PP. Reading through this document, Glow articulates twice – and in all capital letters – that it does not sell any user's personal information. The screenshot in Figure 15, illustrates Glow's attempt to make this statement explicitly clear to users.

Your Privacy and the Use of Your Data.

We recognize the importance of protecting the privacy of our users, particularly given the nature of that information.

You should be aware that Glow may collect and collate a variety of information regarding the use of the Services by you, and may cause the Service to communicate to Glow information about its operations even if you are not connected to the internet or a mobile service. Glow is the sole owner of all such information it collects. Collection and management of such information is governed by our Privacy Policy which is hereby made part of this agreement. Glow uses industry-standard security measures to protect the loss, misuse and alteration of the information under our control. Although we make good faith efforts to store the non-public information uploaded through the Services or collected by Glow in a secure operating environment that is not available to the public, we cannot guarantee complete security. We cannot and do not guarantee that our security measures will prevent third party "hackers" from illegally accessing our site and obtaining access to content or information thereon.

Figure 14. A screen capture of Glow's Privacy Policy with a couple statements highlighted that explain that Glow owns all the data it collects from users through their use of the app.

GLOW PRIVACY POLICY



Last Updated: May 25, 2018

Glow takes issues of privacy and the protection not only of personal health information, but all kinds of information about your private life very seriously. As discussed below we take great efforts to protect your information from disclosure, to use your information only to improve the service for you and to improve our ability to provide services to other people in a similar situation, and to assist researchers in finding better ways to improve health.

WE DO NOT SELL OR RENT YOUR PERSONAL DATA TO THIRD PARTIES. WE DON'T SHARE YOUR INFORMATION (OTHER THAN FORUM POSTS) TO SOCIAL NETWORKS OR OTHER PUBLIC OR SEMI-PUBLIC PLACES UNLESS INSTRUCTED BY YOU TO DO SO.

We only disclose your information to you or as you authorize us to (although we may include certain elements of that information in our data base on an anonymized basis). However we wish to remind you that the kind of information Glow collects is significantly more sensitive than the information you may be providing to other application providers – and you should be extra careful before instructing us to share that information with anyone else. Your menstrual cycles, fertility information and pregnancy status is much more sensitive than the number of steps you took today or where you were when you took your most recent selfie – we recognize that and recommend that you take extra care before deciding to ask us to share it with anyone.

Figure 15. A screen capture of the web version of the Glow application's Privacy Policy, last updated May 25, 2018.

Although written in clear language, two questions come to mind. One, why not make this clear in the process of individuals signing up to become application users? Two, if Glow does not sell a user's data to a third party, why does it maintain ownership over the user's data? These questions, we assert, invite a critique of efficiency and, ultimately, user agency.

Despite the fact that Glow seeks to offer a service for a healthcare demographic they recognize as needing more financial and medical support, their maintaining ownership over individual data so as "to perform and improve our Services, contact you, conduct research, and provide anonymous reporting for medical research purposes" (Terms of Service, 2016) seems suspect. We interpret their decision to maintain ownership over their user's data as a means of generating a revenue under the guise of promoting the app's accessibility and promotion of research. Through the caveat of permitting researchers access to their users' data, Glow forms partnerships with medical companies such as Shady Grove Fertility. This kind of research partnership allows for news stories like "Some infertility programs offer money-back guarantees. Conception may cost \$30,000" (Andrews, 2014, n. p.) to be published and promoted, highlighting how partnerships between Glow and Shady Grove construct a rhetorical validity about the app. Such press coverage benefits from having a physician tout the application in their private practices without critically considering how partnerships like these implicate users of the application. Receiving praise from reliable medical professionals, we concede is not all bad, yet it fails to consider who this app's main stakeholders are: its users. If we reposition Glow's users as important and integral stakeholders, the application should not only allow them to maintain ownership over their data and grant them more transparency with regard to consent, but should also make clear how their data will be used to support the technology company's

financial aims. In the next section, we offer suggestions for a redesign of the application's interface and technical legal documents that would support the repositioning we detail here.

Designing and composing critical feminist action

To fully enact Petersen and Walton's (2018) call for bridging a feminist lens to social justice issues in TPC, we offer a response to our critiques of Glow that evokes critical feminist action for possible users of the fertility and period tracking application as well as other health applications. In what follows, we discuss two TPC areas in which feminist praxis can be deployed to address the gaps and concerns raised in our critique of Glow.

Because Glow's data collection policies erode user agency and, thus, disempower its users, the app functions to be supportive of the company rather than its users. We respond to the egregious lack of user consent by proposing that Glow embrace a more feminist, care-based approach to its efficiency. By doing so, Glow would meet its proclamations that the app promotes user agency throughout the entire process of user interaction with the app. Attention to the entire process in which individuals act as agents over their health and health data through using health applications is important, as Jones reminds us, because it is by attending to the process of how agency emerges that allows for true empowerment. Technical communicators are well suited to make the process of obtaining consent, and the language of what is being consented to, more apparent by redesigning ToS and PP documents to be more transparent. In what follows, we suggest how Glow, and by extension other health applications, may be redesigned to establish more transparent practices for gaining user consent in the access of personal health information. Such a redesign implies a revision of the ToS and PPs for health applications. These documents in their design are often not written in plain language. As technical communicators, we must consider how our design and language may reduce the literacy barriers that are often embedded in these technical documents obtaining user consent.

Designing digital infrastructures for informed consent

As technical communicators work on and with digital platforms, TPC's scholarly scope has shifted to expand beyond the language used to communicate information. TPC has evolved toward considerations that examine the method and design in which information is delivered. For example, Rose's (2016) scholarship on transportation and housing design has inched the field closer to design considerations, pointing out that "the move to digital services makes assumptions about people's digital literacies and access to the Internet and technology. Looking at the resulting designs, we must ask, *'Who, either by intention or by design, is being left out?'*" (p. 428). Her work reiterates that technical communication must consider how language interacts with the dissemination of the information. For Rose, the process to do so begins by resituating TPC from "user-centered" theories towards more "human-centered" approaches. Applying human-centered theories engages the field of TPC with issues of social justice as a focus on the human rather than the user by revealing "the ways that technologies embed existing inequities into their designs and risk further disenfranchising already vulnerable populations" (Rose, 2016, p. 428). In this way, human-centered approaches can support more ethical engagement with their users and facilitate processes that account for agency throughout the user's interaction with the technology, thereby creating a technology capable of empowering its user.

In its current design, however, we find Glow operates through user-centered theories that focus more on the user's "relationship to and use of technology" (Rose, 2016, p. 428). Understanding that human-centered design "has the potential to enlarge the scope of what we pay attention to when designing information, services, and systems" (Rose, 2016, p. 443), we imagine a revision of Glow's interface toward a human-centered design approach. Applying human-centered design to Glow would allow users to engage in a more transparent process of opting in and consenting to Glow's ToS and PP. Such a human-centered approach to Glow would purposefully engage a design that supports user

engagement with its technical documents – evoking on the screen active participation that requires users to read, comprehend, and fully consent to the app’s conditions of use and data collection practices.

Such a redesign of Glow’s interface addresses contemporary issues of informed consent currently facing the field of TPC. For example, Kim, Young, Neimeyer, Baker, and Barfield (2008) suggested that technical communication has not done enough to address emerging ethical challenges of how informed consent is obtained. They point out that prior technical communication scholarship has focused mainly on how language is drafted in informed consent documents. For them, given the increasing complexities of what individuals consent to – particularly in healthcare contexts – technical communicators communicate consent through “not only scientifically accurate and understandable” language but also through “a usable interface” (Kim et al., 2008, p. 336). A redesign of the interface assists in considering and taking up the ethical challenges of designing documents for informed consent. We are hopeful that such a redesign may make users more aware of the data ownership they consented to when using the app and how their personal health data may be further commodified to increase Glow’s prestige and popularity, thus instilling a more consensual enactment of agency over their health and personal information. We contend that some users, given their desire to become pregnant, may weigh these odds and still opt in to using the app. For us, this is a valid, consensual choice as long as they fully understand what it is that they are opting into. Without such acknowledgment, user agency and sense of empowerment over their fertility is not fully exercised.

Revising Terms of Service and privacy policies

Although we see promise in applying principles of human-centered design to Glow to make the process of informed consent more transparent to users, we realize that redesigning the interface for obtaining consent is not enough. Rather, we find technical communicators can engage in critical feminist action by also addressing how ToS and PPs are written. Namely, we need to ensure that if we redesign the interface to create more transparent consent, individuals must be able to comprehend and understand to what they are consenting. As we discussed above, a contract document of this kind fails users by being inaccessible and inaccessibly designed. Such a failure means, at its worst, putting users at risk by making them vulnerable to be further marginalized and exploited through an unequal power dynamic where the technology and company have full authority over users via the control of their data. For this contractual relationship to remain an ethical one, users must have full understanding not only that the app collects their data, but also how the company uses/shares this data.

This ethical dilemma in regard to ToS and PP design must be studied from a risk perspective. Waller, Waller, Haapio, Crag, and Morrisseau (2016) believe that the power imbalance between sellers and consumers can be remedied so long as the sellers provide enough accessible information to their consumers, so that the consumers are able to make “informed decisions about using their products and services” (p. 59). Yet Markel’s (2005) examination of privacy policies revealed that many companies justify the collection of personal information from a utilitarian ethic rather than a right-based view which situates personal information as personal property. From a utilitarian stance, many companies view “personal information as a value to consider when deciding what personal information to collect and what to do with it” (p. 212). A human-centered ethic would, instead, consider the rights and desires of the user over the value such information could provide to the company offering the service.

Markel’s research found that “many privacy policies are written to deprive visitors of their basic rights by exploiting their personal information and that privacy-policy statements employ numerous rhetorical sleights of hand to prevent readers from understanding what happens to their personal information” (p. 197). Providing substantive and clearly defined information about the data a company collects on its users and how that data is aggregated, and subsequently treated, lowers damages incurred because users have a more realistic understanding about their engagement with the service. Users will feel empowered by their choice in using the application because of their

conscious understanding of their role in the engagement with the app. Additionally, in doing so, the company lessens the risk for data breaches and information insecurity. Unfortunately, Glow does not support this kind of user-first practice. Glow, instead, views the information users share through its site as its property by detailing in its ToS and PP that it retains the right to appropriate user data for research, marketing, and improving use of the software. The application retains the right to appropriate user data for medical research, reporting, marketing, and other purposes like perfecting algorithmic design for enhanced tracking of its users.

As such, a more ethical, human-centered redesign of a health application's ToS and PP, like Glow's, can focus on making all facts of data collection transparent, including:

- What data is collected with visual examples of what that data looks like
- How that data is collected also with visuals
- Where that data is stored with specific descriptions of storage vulnerabilities
- What third parties (naming each one explicitly) have access to which parts of user data, with visuals
- How long these third parties have this access and any other relevant stipulations
- What those third parties can do with the data.

Redesigning the application's ToS and PPs to act as a transparent agreement between users and a technology company embraces care as an ethical act of design. In the context of health applications, three options should be enacted to further this new ethical design model. Users should have:

- (1) Options for opting out of any and all of the data collection acts (as indicated above) at any time
- (2) Full right to own all data that they create while using the wearable technology
- (3) The ability to terminate their services, taking all their data with them (the wearable technology will not log or store data upon termination).

By combining these user control options with the redesigned data collection permissions, a health app's ToS and PP become human-centered in ways that enhance transparency and consent. Redesigning these documents for transparency and consent would fulfill the app's promise of user agency and self-empowerment.

Moving towards a more culturally attentive redesign

For us, ethical redesign demands culturally attentive, human-centered design, for it addresses the issue of how users develop agency in their health decisions and remain empowered. Such redesign, we argue, meets TPC social justice objectives that seek to "empower individuals who are disenfranchised and marginalized" (Jones, 2016, p. 345). Also, this redesign addresses Frost's feminist critique of efficiency for how it attempts to reach a more culturally diverse user population by reorienting designers to a more critical awareness of how these technologies collect and use/share user data. Sun's (2012) Culturally Localized User Experience (CLUE) framework offers a "design philosophy that integrates action and meaning in technological design in order to make technology usable and meaningful to culturally diverse users" (p. 267). Being culturally attentive to users in Glow's case would mean recognizing women's right to owning data collected from their bodies. It would mean being attentive to how the composing of ToS and PP documents circumvent user agency and enacting a redesign of those legally binding agreements to be culturally accessible and functional.

Legal documentation design plays a significant role in building an ethical relationship between health applications and users. To support this relationship-building, ToS and PP documents can be more culturally accessible "by replacing the directive nature of legal language with a more invitational tone" and providing "personal contact by phone or meetings" among all stakeholders (Waller

et al., 2016, pp. 55–56). Culturally attentive redesign of health applications, like Glow, benefits the company and its users by minimizing risk and harm that has been, up until now, codified within ToS and PPs. In addition, the technology industry also benefits by establishing a sense of trust between the companies and their users to offer a truly different model of data collection. Such a model adopts care as a central value and practice of company ideology, which supports collaborative and supportive human-centered design.

Implications

This article aims to shed light on an area in need of further examination by feminist and social justice scholars in technical communication: fertility and period tracking applications. Although we critique Glow's ability to provide users with an increased sense of agency over their reproductive health and fertility, we find that Glow does provide a unique and needed service for a health demographic that fails to have affordable access to fertility care. Like Moeller and Frost (2016) assert, "feminists in technical communication need to develop methods for maintaining a difficult balance" (p. 9). As technical communication scholars, we must be critical of technologies that market themselves to vulnerable populations. We see female users, seeking to avoid pregnancy or become pregnant, as a rather vulnerable demographic, particularly in the context of increasing sociopolitical reproductive threats. These fertility and period tracking applications thus provide critical services to those individuals who may not have access to either fertility treatment or pregnancy-related services. We see a critical space, though, for feminist technical communicators and feminist rhetoricians to respond to these applications. Therefore, we must also strike a balance in our critical analysis and "pay attention to the positive aspects and inclusive spirit of feminist work in technical communication even as we carefully interrogate the parts of such work that may be problematic" (Moeller & Frost, 2016, p. 9). Practicing such a balance, we contend, is hard, but doing so works to enact Petersen and Walton's call to bridge feminist critique with social justice action.

As feminist technical communicators, this article identifies a new area of research for feminist rhetoricians and technical communicators committed to improving the technologized spaces of healthcare. Although our analysis focused on one fertility and period tracking application, we contend that there are hundreds of other gendered health applications that could benefit from further analysis and action – not to mention technological applications in general. We find that, as possible users of these applications as well as feminist scholars, we may consider how we can collaborate with health technology companies to make the treatment of personal data more transparent and also more focused on the individual user as a method of care. Such a collaboration can begin with consultation in the design of human-centered application interfaces and policy documents like ToS and PPs that support user agency and empowerment. These technologies, though marketed to assist individuals to improve their access to healthcare information, could benefit from evoking a more apparent feminist commitment to how they claim ownership over their users' data and improve the process whereby they gather users' consent to that data. Frost and Haas' (2017) work opens space in TPC scholarship to consider how we, as consumers, participate in supporting antifeminist technologies and their by-products, like the sonogram. We, too, see this as consumers of these applications, and recognize that one method to begin critical intervention is by identifying socially just approaches to how we, as users, access and decide to consent to using these technologies.

Finally, as teachers of future technical communicators, we invite technical communication classes to examine, critique, and offer attempts at redesign of other health applications. More and more of our student population are users of similar technologies. Inviting our students to become critical technical consumers of these applications, we argue, may assist in producing more critical, ethical, feminist technical writers and designers. Understanding that the data our bodies tell is not a neutral, objective technological act but one that always evokes the technical communicator, is important towards evoking more critical feminist action in these technologies.

Disclosure statement

No potential conflict of interest was reported by the authors.

Notes

1. For the purposes of this article, we focused on the user who wanted to become pregnant. This choice was intentional as we find the exorbitant out-of-pocket costs associated with fertility treatment and alternative family-building, a social justice issue that should be taken up by feminist technical communicators.
2. Scholars have questioned the accessibility of legal technical documents like Terms of Service/Conditions and Privacy Policies. For more, see Vie (2016), Reilly (2016), and Waller et al. (2016).

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